

Building Expressions on the Flow Diagram

Operation vocabulary · Flow diagrams · Word phrases -> expressions

NAME _____

CLASS _____

DATE (YYYY-MM-DD) _____

TEACHER

Mr Smith

OBJECTIVE

0

Learning Intention & Success Criteria

To translate word phrases into algebraic expressions using a flow diagram, and to match operation vocabulary to +, -, $\times$, $\div$.

FOUNDATION

I can identify the coefficient, variable and constant in an algebraic expression (recap from Double 2).

DEVELOPING

I can translate a word phrase into an algebraic expression, using a flow diagram to organise my thinking.

EXTENDING

I can build my own flow diagram and expression for a two-step worded scenario, and explain the order I used.

RECAP

1

Quick Check — Operation Words

Circle the symbol that matches each word.

1. "Product of"



2. "Fewer than"



3. "Shared equally"



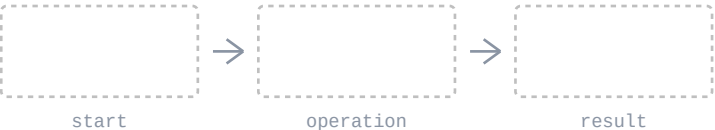
EXPLICIT TEACHING

2

The Flow Diagram & Vocabulary

COPY DOWN: Your teacher will show Slides 4–5. Copy the flow diagram example and the vocabulary table below.

Draw the paperclip pot flow diagram from the board:



Copy the operation vocabulary table — you'll use this all unit:

OPERATION	WORDS THAT MEAN THIS
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Addition (+)	
Subtraction (-)	
Multiplication (×)	
Division (÷)	

3

APPLICATION OF LEARNING

Task 1 — Translate the Phrase

Each question uses a different letter for the unknown amount — read carefully before you write the expression.

FOUNDATION

Word phrase	Expression
There are m marbles in a jar. 5 more marbles are added.	
Sam has p pencils. He gives away 3.	
A number, t , has 6 added to it.	
There are k stickers on a page. 4 are removed.	
A rope is w metres long. It is cut shorter by 2 metres.	
There are c coins in a jar. 10 more are added.	
A number, y , has 7 taken away from it.	
A recipe uses f cups of flour. 2 more cups are added.	

DEVELOPING

Word phrase	Expression
The product of x and 6.	
n divided by 4.	
8 less than a number, d.	
5 subtracted from a number, q.	
Half of a number, r.	
The sum of a number, b, and 9.	
z multiplied by 8.	
A number, h, shared equally among 5 people.	

EXTENDING

Word phrase	Expression
A number, n, is doubled, then 3 is added.	
Twice a number, p, decreased by 4.	
A number, m, has 2 added, then the result is tripled.	

4. Given the expression $4k + 7$, write a worded scenario that matches it.
-
-
5. Given the expression $(t - 5) \div 2$, write a worded scenario that matches it.
-
-

3

APPLICATION OF LEARNING

Task 2 — Complete the Flow Diagram

Fill in whichever boxes are blank — sometimes that's all three, sometimes it's just one.

FOUNDATION

1. "7 added to a number, a."
- →
2. "A number, b, has 4 taken away."
- →
3. "10 more than c."
- →
4. "d decreased by 6."
- →

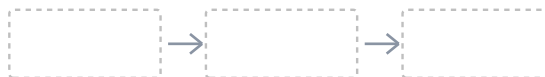
5. "3 added to e."



6. "f minus 9."

**DEVELOPING**

1. Complete the flow diagram for "g divided by 3."



2. Complete the flow diagram for "h multiplied by 5."

3. Given: start = k, operation = $\times 7$. What is the result?4. Given: operation = $+12$, result = $m + 12$. What was the start?

5. Complete the flow diagram for "n shared into 4 equal groups."

6. Given: start = p, result = $p - 8$. What was the operation?**EXTENDING**

1. Draw or describe a two-step flow diagram for "a number, q, is tripled, then 5 is subtracted."

Final expression: _____

2. Draw or describe a two-step flow diagram for "r has 2 added, then the result is doubled."

Final expression: _____

3. A two-step flow diagram: start = s, operation 1 = $+4$, operation 2 = $\times 2$. What is the final expression?

Final expression: _____

APPLICATION OF LEARNING

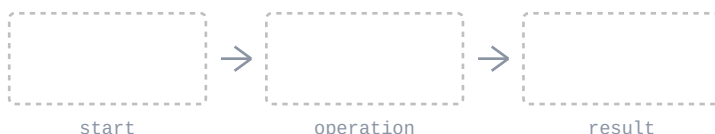
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Task 3 — Build Your Own

One capstone task per tier — pick your tier (or try more than one).

FOUNDATION

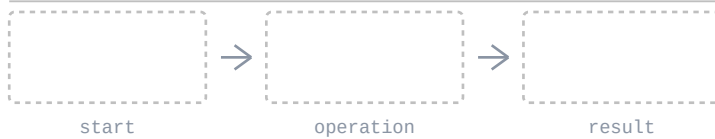
Model this scenario: "5 more than a number, k." Draw the flow diagram, then write the expression.



Expression: _____

DEVELOPING

Write your own one-step worded scenario using multiplication or division, then diagram it.



Expression:

EXTENDING

Chain two operations in one flow diagram (e.g. add, then multiply) for a number, n . Show both steps and the simplified expression.

Final expression:

4 RESPONSIVE TEACHING

4

Quick Check

Your teacher will reveal the class answers — check your thinking against them.

1. True or False: $2x = 6$ is an expression. (circle one)

TRUE**FALSE**

2. Why did you choose that answer? What's different about an expression and an equation?

3. "A box contains apples. Another 2 apples are added. How many apples are there?" was answered with a $+ 2$. What's missing from this answer?

5 REFLECTION

5

Self-Assessment

- ☐ I can identify the coefficient, variable and constant in an algebraic expression.
- ☐ I can translate a word phrase into an algebraic expression, using a flow diagram.
- ☐ I can build my own flow diagram and expression for a two-step worded scenario.

What are you proud of from today?

What do you still need to work on?

6 EXIT TICKET

6

Show What You Know

1. In the expression $9p - 4$, identify the coefficient, variable and constant.

2. Write an expression for: "8 less than m ."

3. Draw or describe a flow diagram for: “a number, n , is doubled, then 3 is added.”

